Campus Water Audit and Conservation Strategy Report 2024

Document ID: DOC_WAT_005 | **Type:** AUDIT_REPORT | **Publication:** 2024-05-18 | **Institution:** VESIT (VESIT Green Campus, Chembur)

CAMPUS WATER AUDIT AND CONSERVATION STRATEGY REPORT 2024

Document Reference: VESIT/AUD/WATER/2024/01

Conducting Cell: Environmental Civil Infrastructure Cell in collaboration with Department of Artificial Intelligence & Data Science (DEP_AIDS).

Facility Scope: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. EXECUTIVE SUMMARY AND STATUTORY CONTEXT

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) encompasses 32,500 m2 of built infrastructure serving a daily population of 3,200 individuals (3,000 students and 200 faculty and administrative staff). In calendar year 2024, the institution conducted a rigorous, meter-level water balance audit across all seven campus buildings: Academic Block A (BLD_ACA), Academic Block B (BLD_ACB), Computer Laboratories (BLD_LAB), Library (BLD_LIB), Auditorium (BLD_AUD), Canteen (BLD_CAN), and Administration Block (BLD_ADM).

The audit established that gross annual freshwater intake from the Brihanmumbai Municipal Corporation (BMC) utility supply reached 48,000,000 litres (48,000 m3). Daily consumption averages 131,500 litres during active semester instruction days. The findings benchmark institutional target TRG_WATER_NEUTRAL, which legally binds the college to curtail municipal piped water demand by 30.0% (saving 14,400,000 litres annually to reach a ceiling of 33,600,000 litres) by 2029, in strict alignment with United Nations SDG 6 (Clean Water and Sanitation) and SDG 12 (Responsible Consumption and Production).

2. END-USE WATER CONSUMPTION BREAKDOWN

Detailed telemetry and fixture log analysis identifies four predominant consumption centers:

2.1 Restrooms, Urinals, and Sanitation Hygiene (46.0% - 22,080,000 litres/year):

Sanitary plumbing across BLD_ACA, BLD_ACB, and BLD_LAB represents the largest campus water draw, consuming an average of 60,500 litres per working day. Conventional flush tanks discharging 10-12 litres per cycle and untuned push-button washbasin taps discharging 9.0 litres/minute account for substantial wastage.

2.2 Canteen and Catering Culinary Facilities (28.0% - 13,440,000 litres/year):

The Canteen building (BLD_CAN) prepares over 2,200 meals and snacks daily. Primary water uses include industrial dishwashing, cooking, utensil pre-rinsing, and reverse osmosis (RO) drinking water purification. Commercial RO purifiers currently exhibit a 65% brine reject rate, discharging approximately 14,000 litres of clean reject water into municipal drains daily.

2.3 Botanical Landscaping and Grounds Irrigation (18.0% - 8,640,000 litres/year):

Maintenance of campus lawns, peripheral tree groves, and the 600 m2 high-density Miyawaki micro-forest (INIT_BIODIVERSITY_ZONE) behind BLD_AUD requires 23,600 litres/day during dry seasons (October through May), currently drawn from municipal tap supply.

2.4 Laboratory Facilities, HVAC Cooling Towers, and Janitorial Cleaning (8.0% - 3,840,000 litres/year):

Includes chemistry preparation, wet lab rinsing, chiller plant cooling tower evaporation makeup, and daily scrubbing of academic corridors.

3. PERFORMANCE OF INTEGRATED RAINWATER HARVESTING SYSTEM (INIT_RWH_SYSTEM)

Under initiative INIT_RWH_SYSTEM, the campus intercepts monsoon precipitation from 18,000 m2 of reinforced concrete terrace catchments across Academic Block A, Academic Block B, and the Library. Runoff is gravity-channeled through leaf screens and multi-media sand-gravel bed filtration units into three underground reinforced concrete storage cisterns with a combined capacity of 450,000 litres.

During the 2024 monsoon season (June to September), Mumbai received 2,420 mm of precipitation. The harvesting system successfully captured, filtered, and utilized 12,400,000 litres of rainwater, directly displacing 25.83% of annual municipal freshwater demand. Harvested rainwater was pumped to dual-plumbing overhead tanks for toilet flushing in BLD_ACA and BLD_ACB, and for cooling tower replenishment. An additional 26,800,000 litres of surplus terrace runoff was routed into four

deep injection borewells, recharging the regional coastal aquifer table and preventing urban storm flooding.

4. ACTION PLAN FOR WATER REDUCTION (TRG_WATER_NEUTRAL)

To achieve the mandated 30% reduction by 2029, the Environmental Infrastructure Cell mandates three capital interventions:

- Comprehensive Aerator Fixture Retrofit: Install pressure-compensating multi-stream aerators on 140 washroom faucets across all academic blocks, restricting flow rates from 9.0 L/min to 3.5 L/min. Projected annual savings: 3,500,000 litres.

- Reverse Osmosis Reject Water Redirection: Plumb the 14,000 L/day RO purification reject line directly into the toilet flushing cisterns of BLD_CAN and BLD_ACA. Projected savings: 3,640,000 litres/year.

- Greywater Reed-Bed Biofilter Upgrade: Construct a decentralized 15,000 L/day horizontal subsurface flow constructed wetland adjacent to BLD_CAN, treating kitchen sink greywater to non-potable standards for landscape and Miyawaki forest irrigation. Projected savings: 3,900,000 litres/year.

Combined savings total 11,040,000 litres annually, ensuring complete fulfillment of TRG_WATER_NEUTRAL.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_WAT_005**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*